

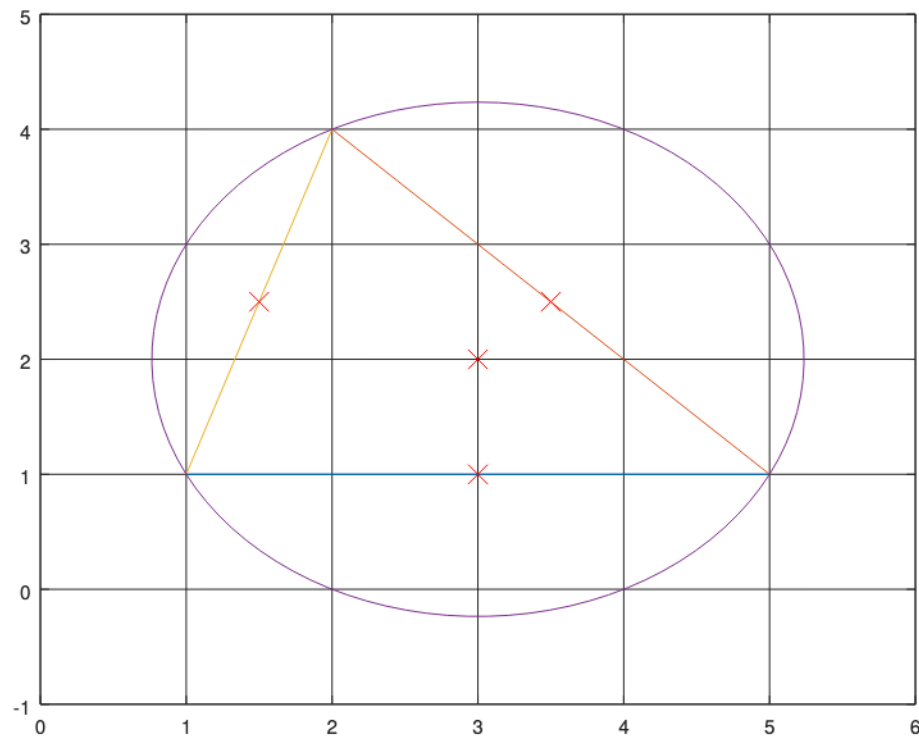
ASSIGN ENT 2 atrix Operations

Tutorial Number: 2
Deadline: October 16 2025
ax. Points: 5
onus

1 Circumcircle

In Matlab/Octave/Numpy matplotlib implement a function which for three input vertices of the triangle in two-dimensional space (expressed by x y coordinates) draws the triangle and its circumscribed circle.

Functions allowed: *plot* and matrix operations



Obrázek 1: Output Example

2 Additional Tasks

Implement the following tasks each as a separate function and demonstrate them on a concrete example:

1. Subtract the smallest element in the corresponding row from each element of the array.
2. Replace all elements in the array that are smaller than the mean of the corresponding column by zero.
3. Find the two most similar line vectors in the matrix (use the cosine of the angle between the vectors as a similarity metric).
4. Mask out the elements of a matrix to get a lower triangular matrix (replace the elements above the diagonal with zeros).

Again use just matrix operations (no loops).